Bridge lemmas for the degree-3 spectral refinement

Repository source: proof/BRIDGE_LEMMAS.md

Contents

Status . . . 1
1. Notation and sourced input . . . 1
2. Tensor-valued harmonic orthogonality . . . 2
   Lemma 2.1 (polarized Bochner identity) . . . 2
   Lemma 2.2 (orthogonality of the curvature tensors) . . . 3
   Corollary 2.3 (projection identity) . . . 4
3. Refined spectral split . . . 4
   Lemma 3.1 . . . 4
4. From the determinant estimate to a nuclear-norm estimate . . . 5
   Lemma 4.1 (two-dimensional nuclear-norm identity) . . . 5
   Lemma 4.2 (integrated nuclear-norm bound for the cubic mode) . . . 5
5. The degree-3 curvature budget . . . 6
   Lemma 5.1 (elementary operator/nuclear duality) . . . 6
   Lemma 5.2 (improved bound on the cubic component) . . . 6
6. Final assembly in the smooth class . . . 7
   Lemma 6.1 (global curvature budget) . . . 7
   Theorem 6.2 (smooth bound from the determinant estimate) . . . 7
   Corollary 6.3 (strict improvement over Nishioka) . . . 7
7. Passage to arbitrary constant-width bodies . . . 8
   Theorem 7.1 . . . 8
8. What this file establishes . . . 8
Source used for the inherited framework . . . 8

Status

This file gives a line-by-line proof of the **noncomputational bridge** from the new degree-3 determinant estimate to the proposed volume bound. The determinant estimate itself is proved in `proof/DEGREE3_TENSOR_IDENTITY.md`; it is isolated as a named input below so that the logical dependency remains explicit.

The argument is for smooth support functions first and then passes to arbitrary convex bodies of constant width by Nishioka's smooth-approximation lemma.

1. Notation and sourced input

Fix a width `d > 0` and put

$$c := \frac{d}{2}.$$

Let `h` be the centered support function of a smooth convex body of constant width `d`, translated so that its degree-one spherical-harmonic component is zero. Thus

$$h(-u) = -h(u), \qquad h \perp \mathcal{H}_1.$$

On the unit sphere S^2, define the self-adjoint tangent endomorphism

$$A_h := \nabla^2 h + hI.$$

Nishioka's support-function formulation gives the pointwise constraint

$$-cI \preceq A_h(u) \preceq cI \qquad (u \in S^2), \tag{1.1}$$

and the volume identity

$$\mathrm{Vol}(K) = \frac{\pi d^3}{6} - \frac{d}{2} E(h), \qquad E(h) := \int_{S^2} \left(\frac{1}{2} |\nabla h|^2 - h^2 \right) d\sigma. \tag{1.2}$$

For tangent endomorphism fields `A` and `B`, write

$$\langle A, B \rangle_2 := \int_{S^2} \mathrm{tr}(AB)\, d\sigma, \qquad \|A\|_2^2 := \langle A, A \rangle_2.$$

Decompose the odd spherical-harmonic expansion as

$$h = h_3 + h_{\geq 5}, \qquad h_{\geq 5} := \sum_{\substack{\ell \geq 5 \\ \ell \text{ odd}}} h_\ell,$$

and set

$$C := A_{h_3}, \qquad N := \|A_h\|_2^2, \qquad N_3 := \|C\|_2^2. \tag{1.3}$$

The proof below uses the following central degree-3 estimate. A human-readable derivation is in `proof/DEGREE3_TENSOR_IDENTITY.md`, and two exact symbolic implementations are in the repository.

Degree-3 determinant estimate. For every $Y \in \mathcal{H}_3$, with $C = A_Y$ and determinant taken on the tangent plane,

$$\|4 \det C\|_2^2 \leq \frac{15183}{17303\pi} \|C\|_2^4. \tag{1.4}$$

Define

$$s := \sqrt{\frac{15183}{17303}}. \tag{1.5}$$

Then (1.4) is equivalently

$$\|4 \det C\|_2 \leq \frac{s}{\sqrt{\pi}} N_3. \tag{1.6}$$

2. Tensor-valued harmonic orthogonality

Lemma 2.1 (polarized Bochner identity)

For smooth real-valued functions f, g on S^2,

$$\int_{S^2} \langle \nabla^2 f, \nabla^2 g \rangle\, d\sigma = \int_{S^2} (\Delta f)(\Delta g)\, d\sigma - \int_{S^2} \langle \nabla f, \nabla g \rangle\, d\sigma. \tag{2.1}$$

Proof Nishioka uses the integrated Bochner identity

$$\int_{S^2} |\nabla^2 q|^2 \, d\sigma = \int_{S^2} (\Delta q)^2 \, d\sigma - \int_{S^2} |\nabla q|^2 \, d\sigma \tag{2.2}$$

for every smooth `q`. Apply (2.2) to `q=f+g` and to `q=f-g`, subtract the second equality from the first, and divide by four. The three quadratic terms polarize respectively to

$$\langle \nabla^2 f, \nabla^2 g \rangle, \qquad (\Delta f)(\Delta g), \qquad \langle \nabla f, \nabla g \rangle.$$

This gives (2.1). □

Lemma 2.2 (orthogonality of the curvature tensors)

If $f \in \mathcal{H}_\ell$ and $g \in \mathcal{H}_m$ with $\ell \neq m$, then

$$\langle A_f, A_g \rangle_2 = 0. \tag{2.3}$$

If instead $f, g \in \mathcal{H}_\ell$, and $\lambda_\ell = \ell(\ell+1)$, then

$$\langle A_f, A_g \rangle_2 = (\lambda_\ell - 1)(\lambda_\ell - 2) \int_{S^2} fg \, d\sigma. \tag{2.4}$$

Proof Expand the pointwise Frobenius product:

$$\begin{aligned} \langle A_f, A_g \rangle &= \langle \nabla^2 f + fI, \nabla^2 g + gI \rangle \\ &= \langle \nabla^2 f, \nabla^2 g \rangle + g \, \Delta f + f \, \Delta g + 2fg, \end{aligned} \tag{2.5}$$

because `tr(I)=2` on the two-dimensional tangent plane. Integrating and using Lemma 2.1 yields

$$\begin{aligned} \langle A_f, A_g \rangle_2 = &\int_{S^2} (\Delta f)(\Delta g) \, d\sigma - \int_{S^2} \langle \nabla f, \nabla g \rangle \, d\sigma \\ &+ \int_{S^2} g \Delta f \, d\sigma + \int_{S^2} f \Delta g \, d\sigma + 2 \int_{S^2} fg \, d\sigma. \end{aligned} \tag{2.6}$$

For spherical harmonics,

$$-\Delta f = \lambda_\ell f, \qquad -\Delta g = \lambda_m g.$$

Also, integration by parts gives

$$\int_{S^2} \langle \nabla f, \nabla g \rangle \, d\sigma = -\int_{S^2} f \Delta g \, d\sigma = \lambda_m \int_{S^2} fg \, d\sigma. \tag{2.7}$$

If $\ell \neq m$, then $\int fg = 0$ by orthogonality of distinct harmonic degrees. Every term on the right of (2.6) is therefore zero, proving (2.3).

If $\ell = m$, write $\lambda = \lambda_\ell$. Substitution in (2.6) gives

$$\begin{aligned} \langle A_f, A_g \rangle_2 &= (\lambda^2 - \lambda - 2\lambda + 2) \int_{S^2} fg \, d\sigma \\ &= (\lambda - 1)(\lambda - 2) \int_{S^2} fg \, d\sigma, \end{aligned}$$

which is (2.4). □

Corollary 2.3 (projection identity)

With the notation (1.3),

$$N = \sum_{\substack{\ell \geq 3 \\ \ell \text{ odd}}} \|A_{h_\ell}\|_2^2, \tag{2.8}$$

and

$$\boxed{N_3 = \langle A_h, C \rangle_2.} \tag{2.9}$$

Proof Because `h` is smooth, its spherical-harmonic expansion converges in every Sobolev norm, in particular in `H^2(S^2)`. Since `f -> A_f = nabla^2 f + fI` is continuous from `H^2` to tensor-valued `L^2`,

$$A_h = \sum_{\substack{\ell \geq 3 \\ \ell \text{ odd}}} A_{h_\ell} \quad \text{in } L^2.$$

Lemma 2.2 makes the summands pairwise orthogonal, proving (2.8). Taking the inner product with `C=A_{h_3}` leaves only the degree-three term:

$$\langle A_h, C \rangle_2 = \langle A_{h_3}, A_{h_3} \rangle_2 = N_3.$$

This proves (2.9). ☐

3. Refined spectral split

Lemma 3.1

For every smooth admissible `h`,

$$\boxed{E(h) \leq \frac{1}{58} N + \frac{9}{319} N_3.} \tag{3.1}$$

Proof Nishioka's harmonic calculation gives, for $h_\ell \in \mathcal{H}_\ell$,

$$E(h_\ell) = \frac{\lambda_\ell - 2}{2} \|h_\ell\|_2^2, \qquad \|A_{h_\ell}\|_2^2 = (\lambda_\ell - 1)(\lambda_\ell - 2)\|h_\ell\|_2^2. \tag{3.2}$$

Hence

$$E(h_\ell) = \frac{1}{2(\lambda_\ell - 1)} \|A_{h_\ell}\|_2^2. \tag{3.3}$$

For degree three, $\lambda_3 = 12$, so the coefficient in (3.3) is `1/22`. For every remaining odd degree $\ell \geq 5$, one has $\lambda_\ell \geq \lambda_5 = 30$, so

$$\frac{1}{2(\lambda_\ell - 1)} \leq \frac{1}{58}. \tag{3.4}$$

Using Corollary 2.3 and orthogonality of `E` across harmonic degrees,

$$\begin{aligned} E(h) &\leq \frac{1}{22} N_3 + \frac{1}{58}(N - N_3) \\ &= \frac{1}{58} N + \left(\frac{1}{22} - \frac{1}{58} \right) N_3. \end{aligned}$$

Finally,

$$\frac{1}{22} - \frac{1}{58} = \frac{36}{1276} = \frac{9}{319},$$

which proves (3.1). □

4. From the determinant estimate to a nuclear-norm estimate

For a self-adjoint endomorphism `M` of a two-dimensional Euclidean space, write `|M|` for its Frobenius norm, `||M||_op` for its operator norm, and `||M||_*` for its nuclear norm.

Lemma 4.1 (two-dimensional nuclear-norm identity)

For every real symmetric `2 x 2` matrix `M`,

$$\boxed{\|M\|_*^2 = |M|^2 + 2|\det M|.} \tag{4.1}$$

Proof Let μ_1, μ_2 be the eigenvalues of `M`. Then

$$\|M\|_* = |\mu_1| + |\mu_2|, \qquad |M|^2 = \mu_1^2 + \mu_2^2, \qquad |\det M| = |\mu_1\mu_2|.$$

Therefore

$$\|M\|_*^2 = (|\mu_1| + |\mu_2|)^2 = \mu_1^2 + \mu_2^2 + 2|\mu_1\mu_2|,$$

which is (4.1). □

Lemma 4.2 (integrated nuclear-norm bound for the cubic mode)

Assume the degree-3 determinant estimate (1.4). Then

$$\boxed{\int_{S^2} \|C(u)\|_*^2 \, d\sigma(u) \le (1+s)N_3,} \tag{4.2}$$

and consequently

$$\boxed{\left(\int_{S^2} \|C(u)\|_* \, d\sigma(u)\right)^2 \le 4\pi(1+s)N_3.} \tag{4.3}$$

Proof Put

$$q(u) := 4\det C(u).$$

By Lemma 4.1,

$$\|C(u)\|_*^2 = |C(u)|^2 + \frac{1}{2}|q(u)|. \tag{4.4}$$

Integrating gives

$$\int_{S^2} \|C\|_*^2 \, d\sigma = N_3 + \frac{1}{2}\|q\|_{L^1}. \tag{4.5}$$

Since $\sigma(S^2) = 4\pi$, Cauchy–Schwarz yields

$$\|q\|_{L^1} \le (4\pi)^{1/2}\|q\|_{L^2} = 2\sqrt{\pi}\,\|q\|_{L^2}. \tag{4.6}$$

Using (1.6),

$$\|q\|_{L^1} \le 2\sqrt{\pi}\left(\frac{s}{\sqrt{\pi}}N_3\right) = 2sN_3. \tag{4.7}$$

Substitution in (4.5) proves (4.2). A second application of Cauchy–Schwarz, now to the nonnegative function `||C||_*`, gives

$$\left(\int_{S^2}\|C\|_*\,d\sigma\right)^2 \le 4\pi\int_{S^2}\|C\|_*^2\,d\sigma.$$

Combining this with (4.2) proves (4.3). □

5. The degree-3 curvature budget

Lemma 5.1 (elementary operator/nuclear duality)

If `A` and `M` are self-adjoint endomorphisms of a two-dimensional Euclidean space, then

$$|\operatorname{tr}(AM)| \le \|A\|_{\mathrm{op}}\|M\|_*. \tag{5.1}$$

Proof Choose an orthonormal eigenbasis e_1, e_2 of `M`, with eigenvalues μ_1, μ_2. Then

$$\operatorname{tr}(AM) = \mu_1\langle Ae_1, e_1\rangle + \mu_2\langle Ae_2, e_2\rangle.$$

Because $|\langle Ae_i, e_i\rangle| \le \|A\|_{\mathrm{op}}$,

$$|\operatorname{tr}(AM)| \le \|A\|_{\mathrm{op}}(|\mu_1| + |\mu_2|) = \|A\|_{\mathrm{op}}\|M\|_*.$$

This proves (5.1). □

Lemma 5.2 (improved bound on the cubic component)

Assume the degree-3 determinant estimate (1.4). Then every smooth admissible `h` satisfies

$$\boxed{N_3 \le \pi d^2(1+s).} \tag{5.2}$$

Proof The pointwise matrix constraint (1.1) implies

$$\|A_h(u)\|_{\mathrm{op}} \le c = \frac{d}{2}. \tag{5.3}$$

By the projection identity (2.9), Lemma 5.1, and (5.3),

$$\begin{aligned} N_3 &= \int_{S^2}\operatorname{tr}(A_hC)\,d\sigma \\ &\le \int_{S^2}|\operatorname{tr}(A_hC)|\,d\sigma \\ &\le c\int_{S^2}\|C\|_*\,d\sigma. \end{aligned} \tag{5.4}$$

If $N_3 = 0$, then (5.2) is immediate. Suppose $N_3 > 0$. Squaring (5.4) and applying (4.3),

$$N_3^2 \le c^2\left(\int_{S^2}\|C\|_*\,d\sigma\right)^2 \le 4\pi c^2(1+s)N_3.$$

Divide by `N_3` and use $c = d/2$:

$$N_3 \le 4\pi \left(\frac{d}{2}\right)^2 (1+s) = \pi d^2 (1+s).$$

This proves (5.2). □

6. Final assembly in the smooth class

Lemma 6.1 (global curvature budget)

Every smooth admissible `h` satisfies

$$\boxed{N \le 2\pi d^2.} \tag{6.1}$$

Proof At each `u`, the two eigenvalues of `A_h(u)` lie in `[-c,c]` by (1.1). Therefore

$$|A_h(u)|^2 \le 2c^2 = \frac{d^2}{2}.$$

Integrating over the sphere of area `4*pi` gives

$$N \le 4\pi \frac{d^2}{2} = 2\pi d^2.$$

□

Theorem 6.2 (smooth bound from the determinant estimate)

Every smooth convex body `K` of constant width `d` satisfies

$$\boxed{\mathrm{Vol}(K) \ge \frac{\pi}{1914}\left(259 - 27\sqrt{\frac{15183}{17303}}\right) d^3.} \tag{6.2}$$

Proof Combine Lemma 3.1, Lemma 5.2, and Lemma 6.1:

$$\begin{aligned} E(h) &\le \frac{1}{58}N + \frac{9}{319}N_3 \\ &\le \frac{2\pi d^2}{58} + \frac{9}{319}\pi d^2(1+s) \\ &= \frac{\pi d^2}{319}(20+9s). \end{aligned} \tag{6.3}$$

Insert (6.3) in the volume identity (1.2):

$$\begin{aligned} \mathrm{Vol}(K) &\ge \frac{\pi d^3}{6} - \frac{d}{2}\frac{\pi d^2}{319}(20+9s) \\ &= \pi d^3 \left(\frac{1}{6} - \frac{20+9s}{638}\right) \\ &= \frac{\pi d^3}{1914}(259 - 27s). \end{aligned}$$

Substitute the definition (1.5) of `s` to obtain (6.2). □

Corollary 6.3 (strict improvement over Nishioka)

The coefficient in (6.2) is strictly larger than `4*pi/33`.

Proof Since `15183 < 17303`, one has `s < 1`. Direct subtraction gives

$$\frac{1}{1914}(259 - 27s) - \frac{4}{33} = \frac{9}{638}(1 - s) > 0.$$

Multiplying by `pi` proves the claim. □

7. Passage to arbitrary constant-width bodies

Theorem 7.1

Together with the degree-3 determinant estimate proved in `proof/DEGREE3_TENSOR_IDENTITY.md`, equation (6.2) holds for every three-dimensional convex body of constant width `d`, without a smoothness assumption.

Proof Let `K` be an arbitrary convex body of constant width `d`. Nishioka's Lemma 2 states that `K` can be approximated in the Hausdorff metric by convex bodies `K_j` of the same constant width with smooth support functions, and that volume is continuous for Hausdorff convergence. Thus

$$K_j \longrightarrow K \quad \text{and} \quad \mathrm{Vol}(K_j) \longrightarrow \mathrm{Vol}(K).$$

Theorem 6.2 applies to every `K_j`, so

$$\mathrm{Vol}(K_j) \geq \frac{\pi}{1914}\left(259 - 27\sqrt{\frac{15183}{17303}}\right) d^3.$$

Pass to the limit as `j -> infinity`. The right-hand side is independent of `j`, and volume convergence gives the same inequality for `K`. □

8. What this file establishes

This file gives the complete bridge from Nishioka's support-function framework to the proposed volume coefficient. Its only non-sourced mathematical input is the degree-3 determinant estimate (1.4), which is proved in `proof/DEGREE3_TENSOR_IDENTITY.md` using the same tensor and determinant conventions tested by the Python and R programs.

The two proof files have now been merged into the publication-style manuscript `paper/degree3_spectral_refinement.tex`. The internal line-by-line audit is recorded in `proof/PROOF_AUDIT.md`. External expert review remains open.

Source used for the inherited framework

Akatsuki Nishioka, "An improved lower bound for the three-dimensional Blaschke–Lebesgue problem from spectral and dual perspectives," Journal of Mathematical Analysis and Applications 566 (2027), 131038, especially Sections 2–4 and Lemma 2.